

Capítulo 12

Chapter Precap

- Introducimos modelos para predecir variables categóricas con más de dos categorías.
- Discutimos el uso de modelos multinomiales para modelar respuestas, hacer predicciones y hacer mapas territoriales.
- A continuación, se discute la regresión logística ordinal para variables categóricas ordenadas.

Multinomial Distribution

- Una distribución de tres parámetros para resultados categoricos.

- 1 n : The number of trials, a positive integer.
- 2 J : The number of possible outcome categories, a positive integer.
- 3 $p_1, \dots, p_J$: A vector of probabilities of observing each of the J outcomes where $\sum p_j = 1$.

- La variable es un vector de recuentos para cada categoría:

$$y_1, \dots, y_J \sim \text{multinomial}(p_1, \dots, p_J, n)$$

Multinomial Data

```
rmultinom (10, 1, c(.4, .1, .2, .3))
```

```
##           [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]  
## [1,]         1    0    0    1    0    0    0    0    0    0  
## [2,]         0    0    1    0    0    1    0    0    0    0  
## [3,]         0    1    0    0    1    0    0    0    1    0  
## [4,]         0    0    0    0    0    0    1    1    0    1
```

```
rowMeans (rmultinom (10000, 1, c(.4, .1, .2, .3)))
```

```
## [1] 0.3985 0.1020 0.1987 0.3008
```

Multinomial Data

```
set.seed (1)
multinomial_variable = rmultinom (10, 100, c(.4, .1, .2, .3))
multinomial_variable
##           [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [1,]      42   37   41   41   41   37   36   47   31   37
## [2,]      10   15    7   14    8   14   13   10   14   12
## [3,]      24   21   18   15   19   18   22   16   22   22
## [4,]      24   27   34   30   32   31   29   27   33   29

# the row means closely approximate p
rowMeans(multinomial_variable)/100
## [1] 0.390 0.117 0.197 0.296
```

Multinomial Regression

- Dado:
 - Vectores de recuentos para cada categoría: $[y_1, y_2, \dots, y_J]$
 - Y el total de observaciones dentro de cada vector, n .
- Predice la probabilidad de observar cada categoría, $[y_1, y_2, \dots, y_J]$, en función de una combinación de predictores.

$$y_1, \dots, y_J \sim \text{multinomial}(p_1, \dots, p_J, n)$$

Multinomial Logits and the Softmax Function

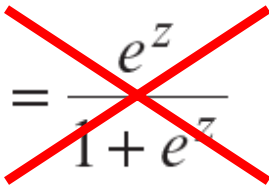
- Función de enlace logístico

The diagram illustrates the logistic link function with the following components and annotations:

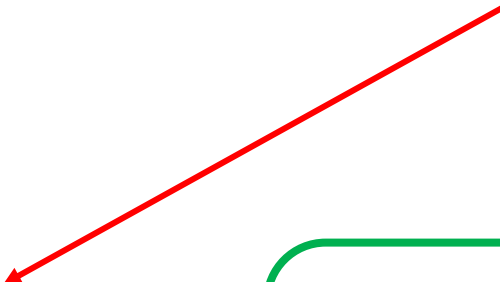
- Equation:**
$$P(y = 1) = \frac{\text{success}}{\text{failure} + \text{success}} = \frac{e^z}{1 + e^z}$$
- Annotations:**
 - Logaritmo de cantidad de éxitos esperados:** Points to the e^z term in the numerator.
 - Cantidad de éxitos esperados:** Points to the e^z term in the numerator.
 - Probabilidad:** Points to the $P(y = 1)$ term.
 - Cuotas:** Points to the $\text{failure} + \text{success}$ term in the denominator.
 - Fracasos esperados = 1:** Points to the 1 term in the denominator.

Multinomial Logits and the Softmax Function

- Para $J > 2$ categorías usamos la función 'softmax':

$$P(y = 1) = \frac{\text{success}}{\text{failure} + \text{success}} = \frac{e^z}{1 + e^z}$$


$$P(Y = j) = \frac{e^{z_j}}{\sum_{j=1}^J e^{z_j}}$$


$$P(Y = j) = \frac{e^{z_j}}{1 + \sum_{j=2}^J e^{z_j}}$$



Multinomial Regression

- Predecimos los valores de z_j para las categorías 2-J utilizando una ecuación de predicción.

$$P(Y = j) = \frac{e^{z_j}}{1 + \sum_{j=2}^J e^{z_j}}$$

$$z_j = \beta_j + \beta_{j1} \cdot x_1 + \beta_{j2} \cdot x_2 + \dots + \beta_{jk} \cdot x_k$$

- z_1 se fija en 0 y también sus parámetros.

$$z_1 = \alpha_1 + \beta_{11} \cdot x_1 + \beta_{12} \cdot x_2 + \dots + \beta_{1k} \cdot x_k$$

$$z_1 = 0 + 0 \cdot x_1 + 0 \cdot x_2 + \dots + 0 \cdot x_k$$

$$z_1 = 0$$

What is z_j ?

- En la regresión logística, z era un logit.
$$P(Y = j) = \frac{e^{z_j}}{1 + \sum_{j=2}^J e^{z_j}}$$
- Podríamos llamarlos 'logits multinomiales'. Es más común que se les llame la “puntaje” para cada categoría.
- e^j es la cuenta de categoría j si la cuenta de categoría 1 es 1.
- También se puede entender como una variable latente.

Latent Variables

- Variables latentes: Observadas matemáticamente pero no observadas directamente.
- Cuando predecimos el género percibido, podemos interpretar z como la feminidad/masculinidad subyacente que conduce a una clasificación binaria.
- Para las categorías niño, niña, hombre, mujer, nuestra variable latente es una medida continua de cuánto se parece una voz a cada categoría.

Latent Variables

As a latent-variable model [\[edit \]](#)

The logistic model has an equivalent formulation as a [latent-variable model](#). This formulation is common in the theory of [discrete choice](#) models and makes it easier to extend to certain more complicated models with multiple, correlated choices, as well as to compare logistic regression to the closely related [probit model](#).

Imagine that, for each trial i , there is a continuous [latent variable](#) Y_i^* (i.e. an unobserved [random variable](#)) that is distributed as follows:

$$Y_i^* = \beta \cdot \mathbf{X}_i + \varepsilon_i$$

where

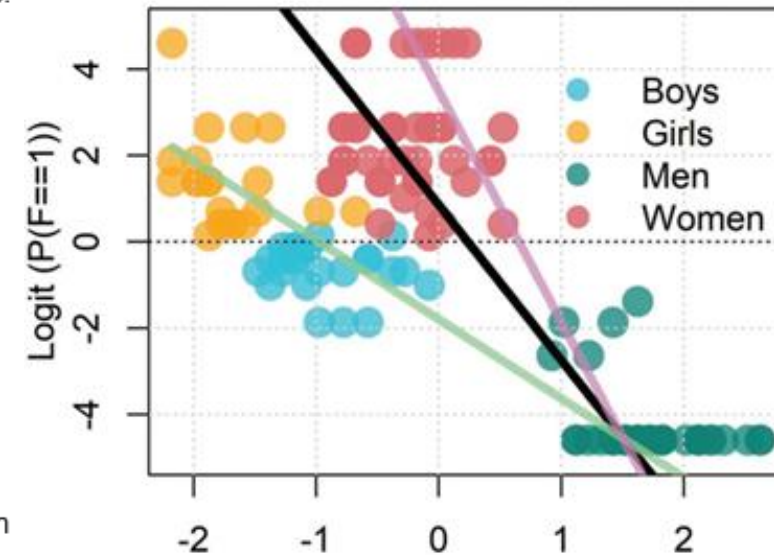
$$\varepsilon_i \sim \text{Logistic}(0, 1)$$

i.e. the latent variable can be written directly in terms of the linear predictor function and an additive random [error variable](#) that is distributed according to a standard [logistic distribution](#).

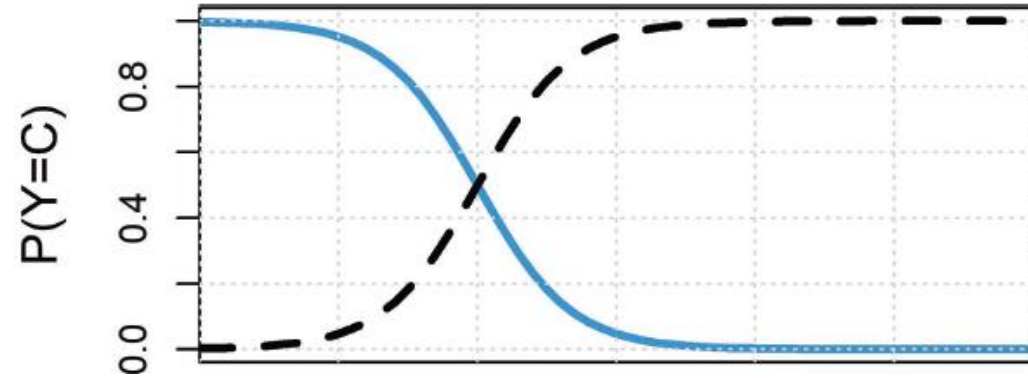
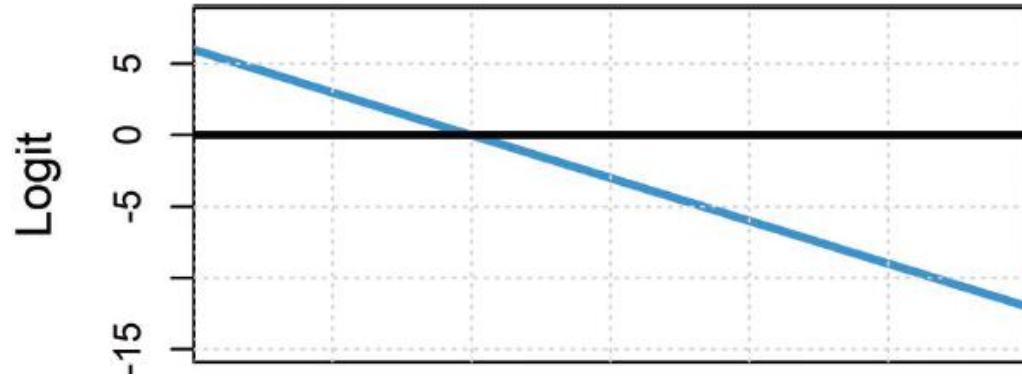
Then Y_i can be viewed as an indicator for whether this latent variable is positive:

$$Y_i = \begin{cases} 1 & \text{if } Y_i^* > 0 \text{ i.e. } -\varepsilon_i < \beta \cdot \mathbf{X}_i, \\ 0 & \text{otherwise.} \end{cases}$$

The choice of modeling the error variable specifically with a standard logistic distribution, rather than a general logistic distribution with the location and scale set to arbitrary values, seems restrictive, but in fact, it is not. It must be kept in mind that we can choose the regression coefficients ourselves, and very often can use them to offset changes in the parameters of the error variable's distribution. For example, a logistic error-variable distribution with a non-zero location parameter μ (which sets the mean) is equivalent to a distribution with a zero location parameter, where μ has been added to the intercept coefficient. Both situations produce the same value for Y_i^* regardless of settings of explanatory variables. Similarly, an arbitrary scale parameter s is equivalent to setting the scale parameter to 1 and then dividing all regression coefficients by s . In the latter case, the resulting value of Y_i^* will be smaller by a factor of s than in the former case, for all sets of explanatory variables — but critically, it will always remain on the same side of 0, and hence lead to the same Y_i choice.



Comparison to Logistic Regression

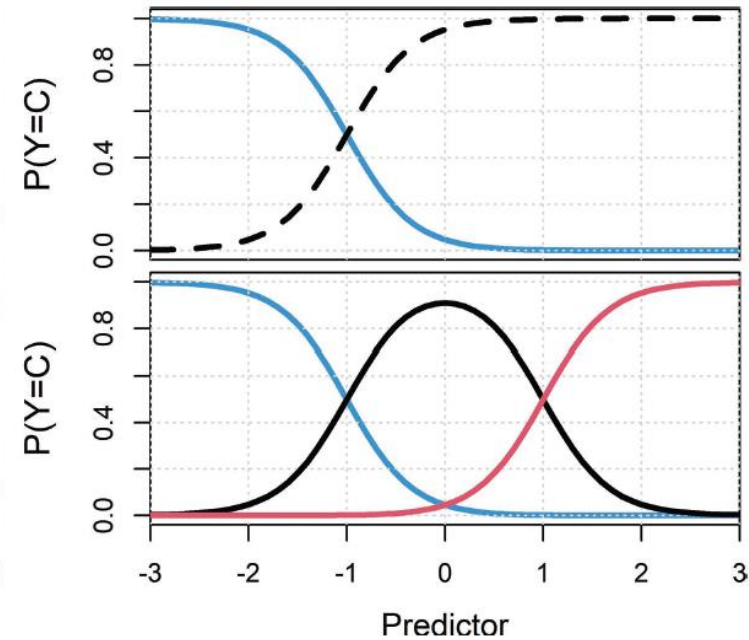
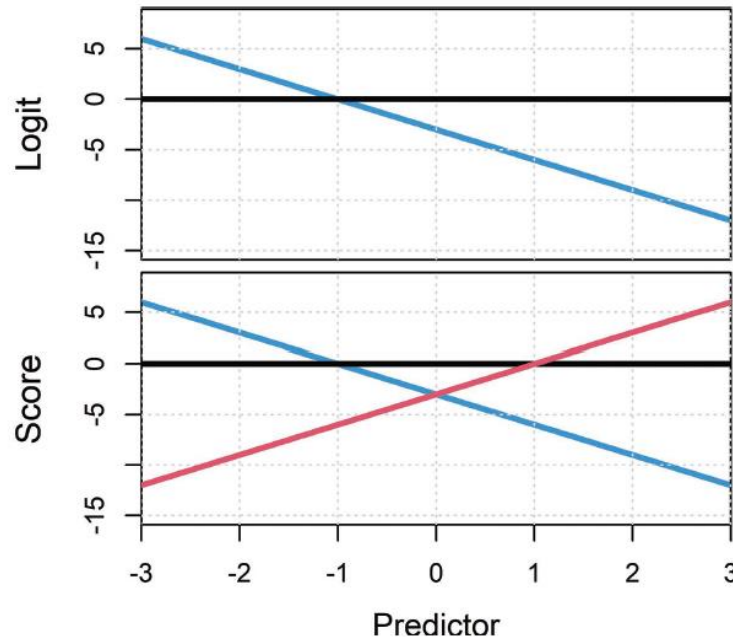


$$z_1 = 0 + x \cdot 0$$
$$z_2 = -3 + x \cdot -3$$

$$P(Y = 2) = \frac{e^{z_2}}{e^{z_1} + e^{z_2}} = \frac{e^{z_2}}{1 + e^{z_2}}$$

$$P(Y = 1) = \frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{z_2}}$$

Comparison to Logistic Regression



$$z_1 = 0 + x \cdot 0$$

$$z_2 = -3 + x \cdot -3$$

$$z_3 = -3 + x \cdot 3$$

$$P(Y = 3) = \frac{e^{z_3}}{1 + e^{z_2} + e^{z_3}}$$

$$P(Y = 2) = \frac{e^{z_2}}{1 + e^{z_2} + e^{z_3}}$$

$$P(Y = 1) = \frac{1}{1 + e^{z_2} + e^{z_3}}$$

Data and Research Questions

(Q1) Can we use speaker f0 and VTL to predict apparent speaker category?

```
library (brms)
library (bmmb)
data (exp_data)

# new dependent variable
exp_data$y = cbind(b = as.numeric(exp_data$C=='b'),
                  g = as.numeric(exp_data$C=='g'),
                  m = as.numeric(exp_data$C=='m'),
                  w = as.numeric(exp_data$C=='w'))

# variable representing the size (n) of each observation.
exp_data$size = 1

# preparation of quantitative predictors
# as in previous chapters
exp_data$vtl_original = exp_data$vtl
exp_data$vtl = exp_data$vtl - mean (exp_data$vtl)

exp_data$f0_original = exp_data$f0
exp_data$f0 = exp_data$f0 - mean(exp_data$f0)
exp_data$f0 = exp_data$f0 / 100
```

Data and Research Questions

```
# first 6 elements of dependent variable
```

```
head (exp_data$y)
```

```
##           b g m w
```

```
## [1,] 0 1 0 0
```

```
## [2,] 0 0 0 1
```

```
## [3,] 0 1 0 0
```

```
## [4,] 0 1 0 0
```

```
## [5,] 1 0 0 0
```

```
## [6,] 1 0 0 0
```

```
# first 6 elements of apparent speaker category factor
```

```
head (exp_data$C)
```

```
## [1] "g" "w" "g" "g" "b" "b"
```

Description of our Model

```
y|trials(size) ~ vt1+f0 + (vt1+f0|x|L) + (1|y|S)
```



```
y_1 ~ vt1+f0 + (vt1+f0|x|L) + (1|y|S)  
y_2 ~ vt1+f0 + (vt1+f0|x|L) + (1|y|S)  
y_3 ~ vt1+f0 + (vt1+f0|x|L) + (1|y|S)  
y_4 ~ vt1+f0 + (vt1+f0|x|L) + (1|y|S)
```

Description of our Model

$$y_1[i], y_2[i], y_3[i], y_4[i] \sim \text{multinomial}(p_1[i], p_2[i], p_3[i], p_4[i], \text{size}[i])$$

for $j = 1, \dots, 4$

$$p_{j[i]} = \frac{e^{z_{j[i]}}}{\sum_{j=1}^J e^{z_{j[i]}}}$$

$$\begin{aligned} z_{j[i]} &= a_j + b_{j[i]} \cdot \text{vtl}[i] + c_{j[i]} \cdot \text{f0}[i] \\ a_{j[i]} &= \text{Intercept}_j + L_{j[L[i]]} + S_{j[S[i]]} \\ b_{j[i]} &= VTL_j + VTL_j : L_{j[L[i]]} \\ c_{j[i]} &= F0_j + F0_j : L_{j[L[i]]} \end{aligned}$$

for $j = 2, \dots, 4$

$$\text{Intercept}_j \sim t(3, 0, 3)$$

$$VTL_j, F0_j \sim t(3, 0, 3)$$

$$\sigma_{L_j}, \sigma_{VTL_j:L_j}, \sigma_{F0_j:L_j}, \sigma_{S_j} \sim t(3, 0, 3)$$

$$\begin{bmatrix} S_2[\cdot] \\ S_3[\cdot] \\ S_4[\cdot] \end{bmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma_S \right)$$

$$\begin{bmatrix} L_2[\cdot] \\ VTL_2 : L_2[\cdot] \\ F0_2 : L_2[\cdot] \\ L_3[\cdot] \\ VTL_3 : L_3[\cdot] \\ F0_3 : L_3[\cdot] \\ L_4[\cdot] \\ VTL_4 : L_4[\cdot] \\ F0_4 : L_4[\cdot] \end{bmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma_L \right)$$

$$R_S \sim \text{LKJCorr}(2)$$

$$R_L \sim \text{LKJCorr}(2)$$

Description of our Model

Our vector of counts $(y_{1[i]}, y_{2[i]}, y_{3[i]}, y_{4[i]})$ is assumed to come from a multinomial distribution with unknown parameters $p_{1[i]}, p_{2[i]}, p_{3[i]}, p_{4[i]}$, and an n equal to 1 (size) for all trials. For each of our j response categories, the probability of observing that outcome on a trial i , $p_{j[i]}$, is found by combining the score for each category for that trial ($z_{j[i]}$) using the softmax link function. The score for a trial is equal to a trial-dependent combination of an intercept, an effect for VTL, and an effect for f_0 . The intercept for category j for trial i ($a_{j[i]}$) varies according to an overall model intercept for that category (Intercept_j) and speaker ($S_j[S[i]]$) and listener-dependent ($L_j[L[i]]$) variations from this. The VTL slope for category j for trial i ($b_{j[i]}$) varies according to an overall VTL slope for that category (VTL_j) and listener-dependent ($\text{VTL} : L_j[L[i]]$) variations from this. The f_0 slope for category j for trial i ($c_{j[i]}$) varies according to an overall f_0 slope for that category (F0_j) and listener-dependent ($\text{F0} : L_j[L[i]]$) variations from this.

Fitting the Model

```
multinomial_prior =  
  c(set_prior("student_t(3, 0, 3)", class = "Intercept", dpar="mug"),  
    set_prior("student_t(3, 0, 3)", class = "b", dpar="mug"),  
    set_prior("student_t(3, 0, 3)", class = "sd", dpar="mug"),  
    set_prior("student_t(3, 0, 3)", class = "Intercept", dpar="mum"),  
    set_prior("student_t(3, 0, 3)", class = "b", dpar="mum"),  
    set_prior("student_t(3, 0, 3)", class = "sd", dpar="mum"),  
    set_prior("student_t(3, 0, 3)", class = "Intercept", dpar="muw"),  
    set_prior("student_t(3, 0, 3)", class = "b", dpar="muw"),  
    set_prior("student_t(3, 0, 3)", class = "sd", dpar="muw"),  
    set_prior("lkj_corr_cholesky (2)", class = "cor"))
```

```
# Fit the model yourself
```

```
model_multinomial =  
  brms::brm (y|trials(size) ~ vtl+f0 + (vtl+f0|x|L) + (1|y|S),  
            data=exp_data, family="multinomial", chains=4, cores=4,  
            warmup=1000, iter = 5000, thin = 4, prior = multinomial_prior)
```

Interpreting the Model

```
brms::fixef (model_multinomial)
```

##		<i>Estimate</i>	<i>Est.Error</i>	<i>Q2.5</i>	<i>Q97.5</i>	
##	mug_Intercept	-2.4516	0.4508	-3.3802	-1.6018	←
##	mum_Intercept	-2.1677	0.4671	-3.1804	-1.3566	←
##	muw_Intercept	-0.6317	0.4336	-1.5063	0.2166	←
##	mug_vt1	-2.0876	0.3746	-2.8648	-1.3766	
##	mug_f0	1.4335	0.6987	0.1241	2.8640	
##	mum_vt1	3.0820	0.5938	2.0761	4.3871	
##	mum_f0	-2.6146	1.0238	-4.7816	-0.7190	
##	muw_vt1	0.9464	0.4613	0.0266	1.8643	
##	muw_f0	2.2001	1.0943	0.1476	4.4024	

Predicting Responses

```
multi_pred_re = fitted (model_multinomial)
```

```
head(multi_pred_re[,1,])
```

```
##           P(Y = b) P(Y = g)  P(Y = m) P(Y = w)
## [1,]    0.2921    0.6190 0.0017775  0.08714
## [2,]    0.2104    0.6100 0.0014059  0.17815
## [3,]    0.2804    0.6624 0.0016820  0.05551
## [4,]    0.4052 → 0.4507 0.0024148  0.14168
## [5,]    0.4385    0.5003 0.0036467  0.05764
## [6,]    0.2218    0.6879 0.0003084  0.09000
```

```
# find highest posterior probability from each category
```

```
predicted = apply (multi_pred_re[,1,],1,which.max)
```

```
head (predicted)
```

```
## [1] 2 2 2 2 2 2
```

```
# use modal category to get a category label
```

```
predicted_category = c("b","g","m","w")[predicted]
```

```
head (predicted_category)
```

```
## [1] "g" "g" "g" "g" "g" "g"
```



Predicting Responses

```
table (observed_category = exp_data$C, predicted_category)
##
##      predicted_category
## observed_category    b     g     m     w
##      b 269    78    27    67
##      g  74  290     0    62
##      m   5   1  624     5
##      w  30  19   9  525
```

```
# overall correct
mean (predicted_category == exp_data$C)
## [1] 0.8192

# correct predictions by category
tab = xtabs (~ exp_data$C + predicted_category)
diag(tab) / rowSums(tab)
##      b      g      m      w
## 0.6100 0.6808 0.9827 0.9005
```

Category Boundaries

- Los planos no se intersecan o se intersecan en una línea. Si se intersecan en una línea, esto se puede encontrar así:

$$z_1 = a + b \cdot x + c \cdot y$$

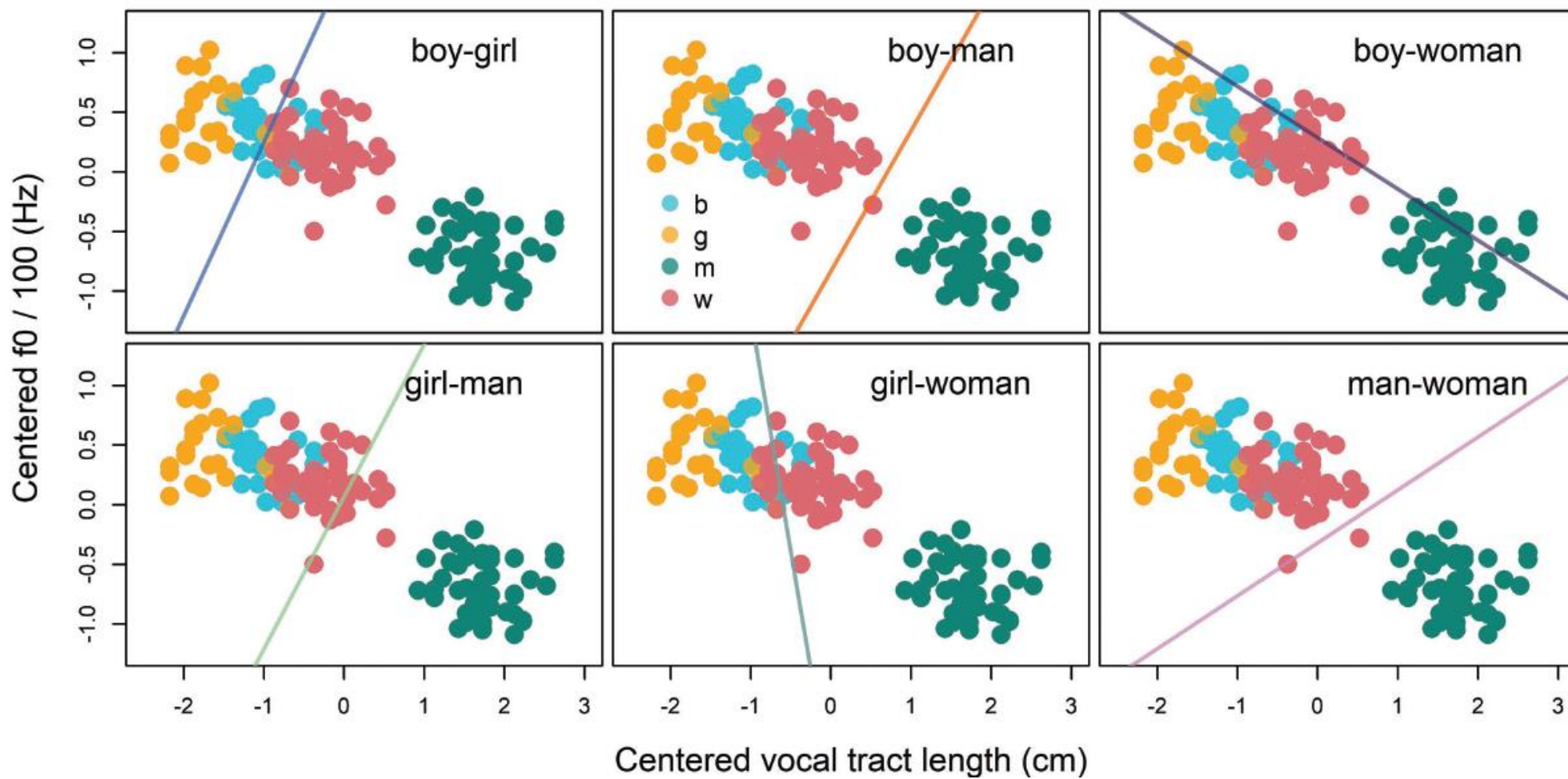
$$z_2 = d + e \cdot x + f \cdot y$$

$$z_1 = z_2$$

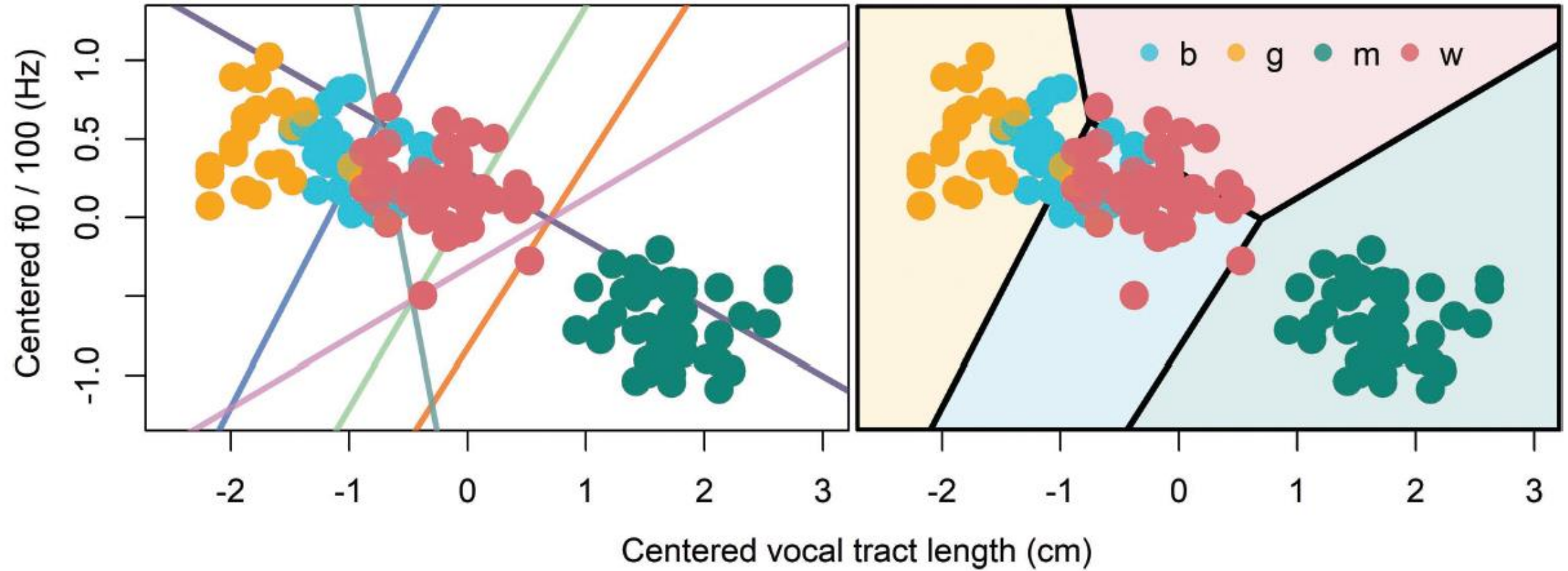

$$a + b \cdot x + c \cdot y = d + e \cdot x + f \cdot y$$


$$y = \frac{-a + d}{c - f} + \frac{-b + e}{c - f} \cdot x$$

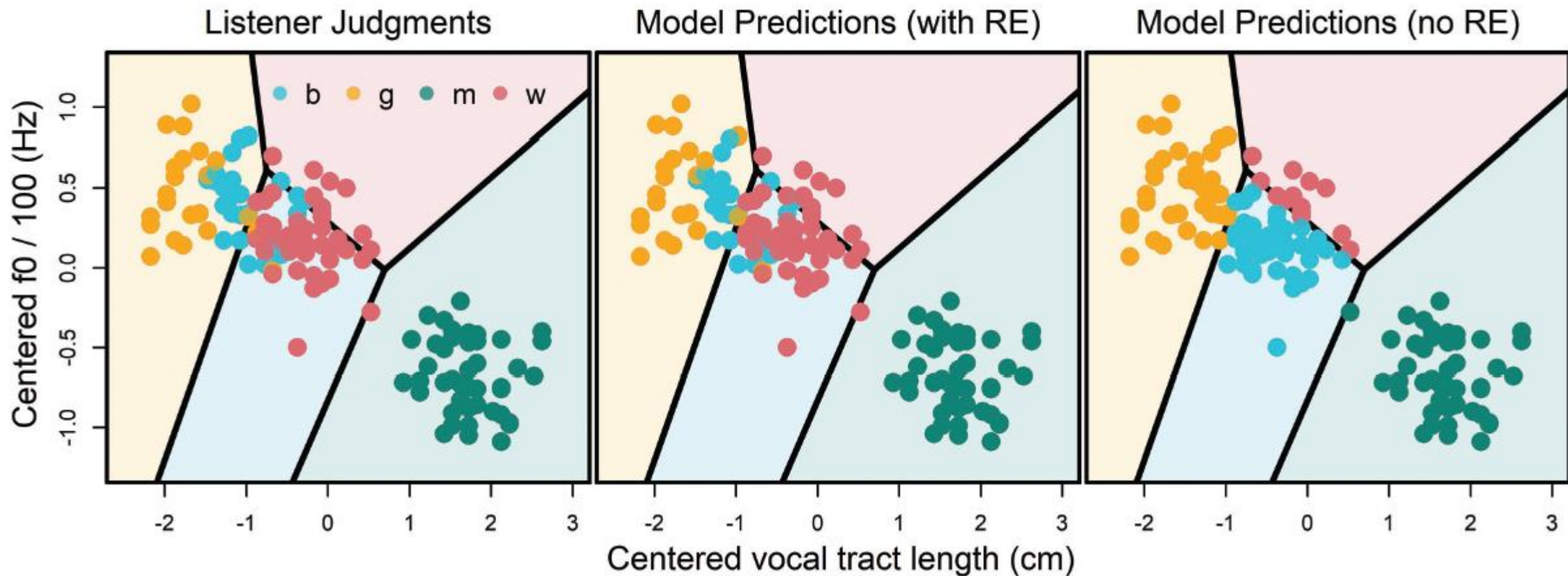
Category Boundaries



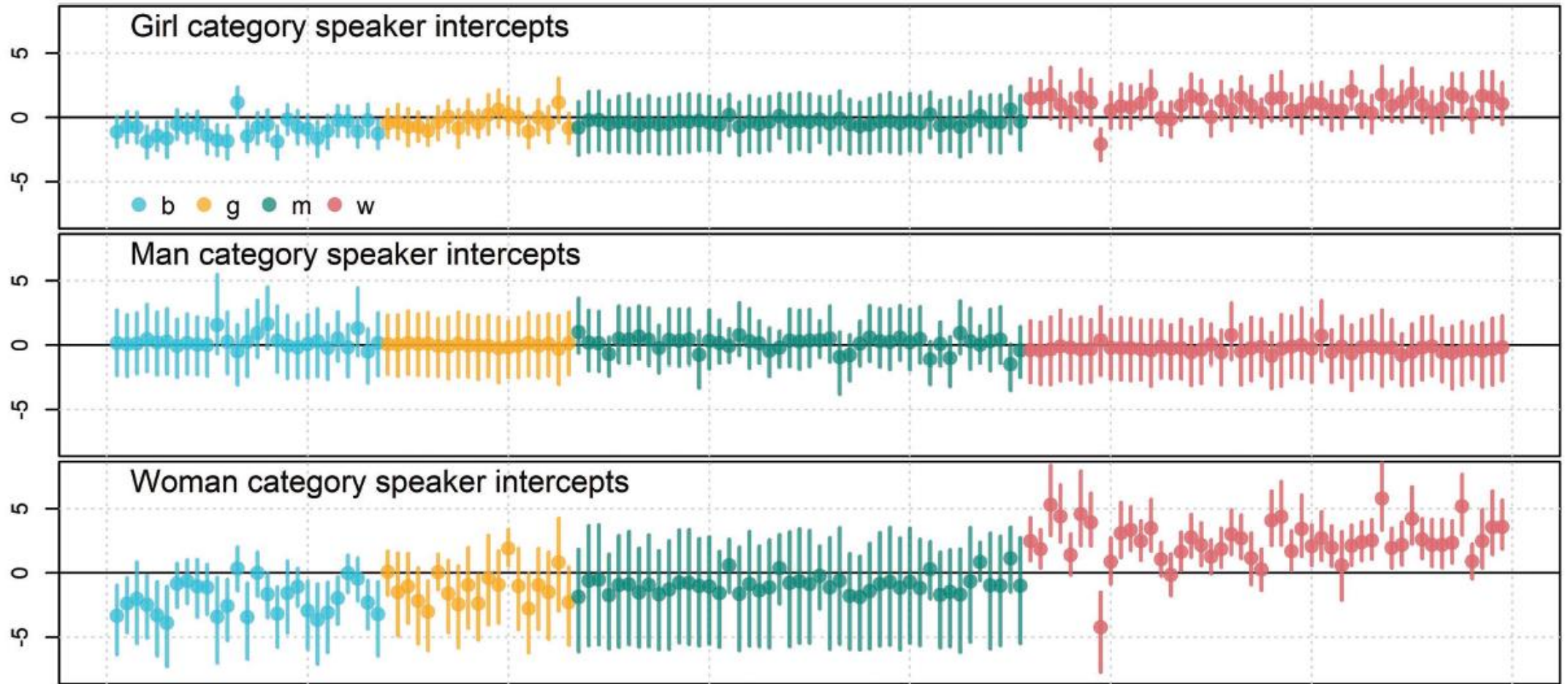
Territorial Maps



Territorial Maps

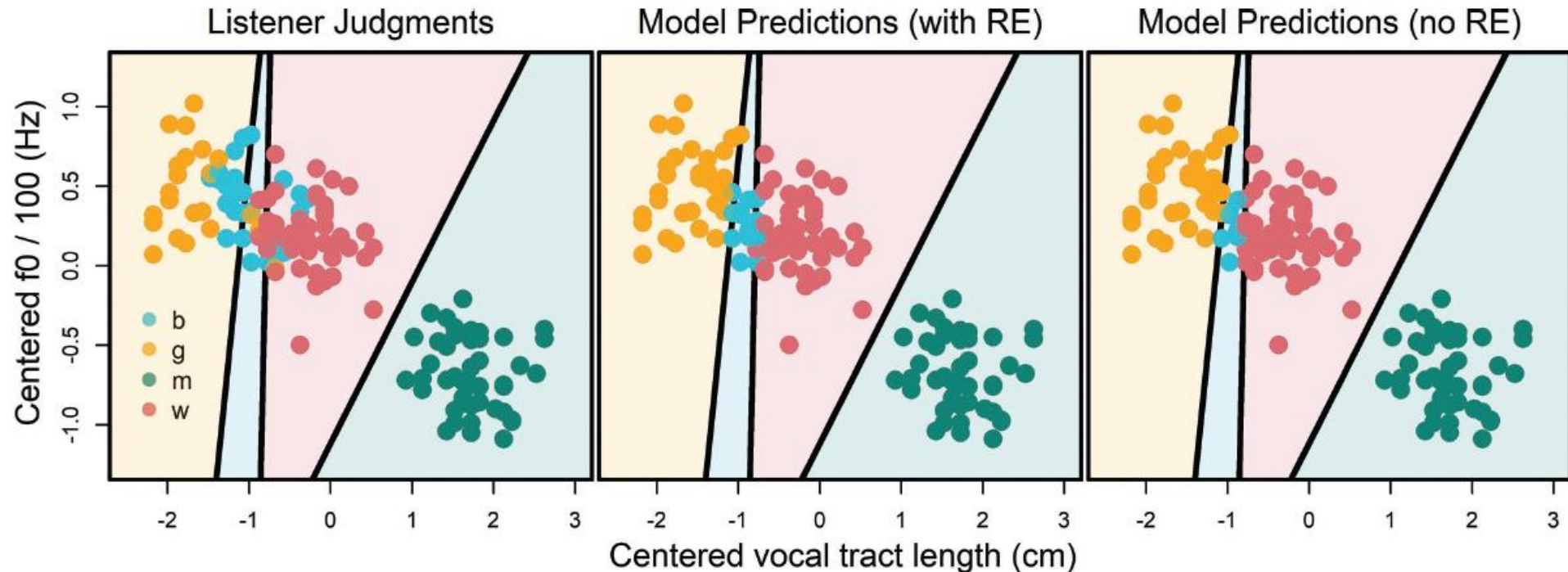


Speaker Effects

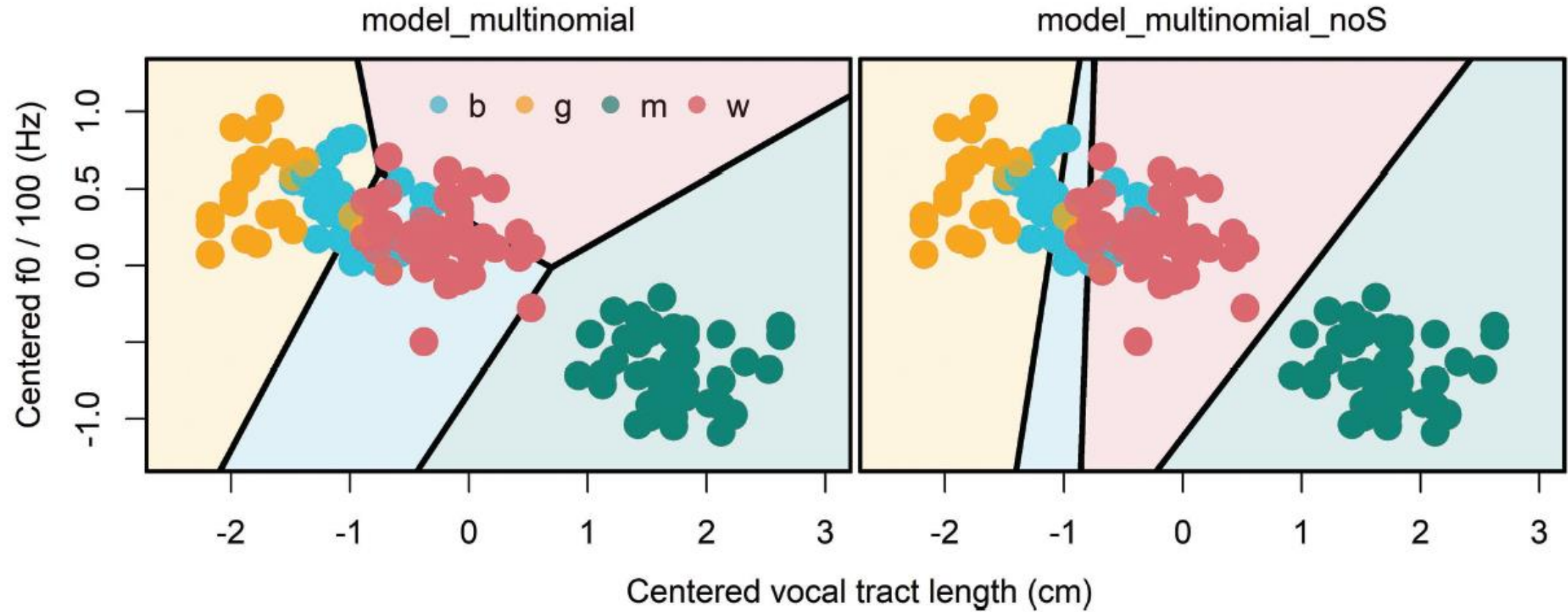


No Speaker Effects

```
# Fit the model yourself
model_multinomial_noS =
  brms::brm (y|trials(size) ~ vtl+f0 + (vtl+f0|x|L), data=exp_data,
            family=multinomial(), chains=4, cores=4, warmup=1000,
            iter = 5000, thin = 4, prior = multinomial_prior)
```



Effect of Speaker Effects



Ordinal Regression

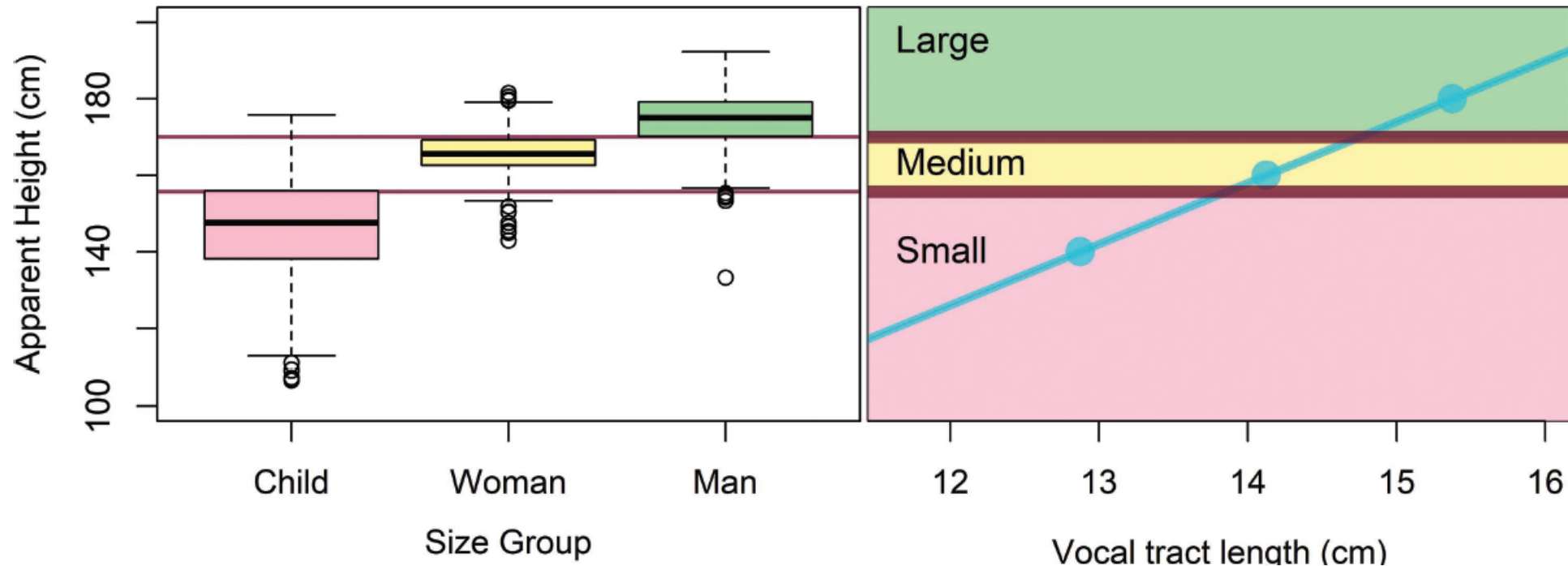
- La regresión multinomial es una extensión de la regresión logística.
- La regresión ordinal es un poco más extraña.
- Sin embargo, muchos datos lingüísticos (por ejemplo, datos de escala Likert) requieren una regresión ordinal.

Ordinal Data

- Necesitas una regresión ordinal si la variable de respuesta:
 - Tiene un orden inherente (es decir, no es intercambiable).
 - Pero no es estrictamente cuantitativo.
 - Tiene un número discreto de resultados.
 - Tiene un número pequeño de resultados.

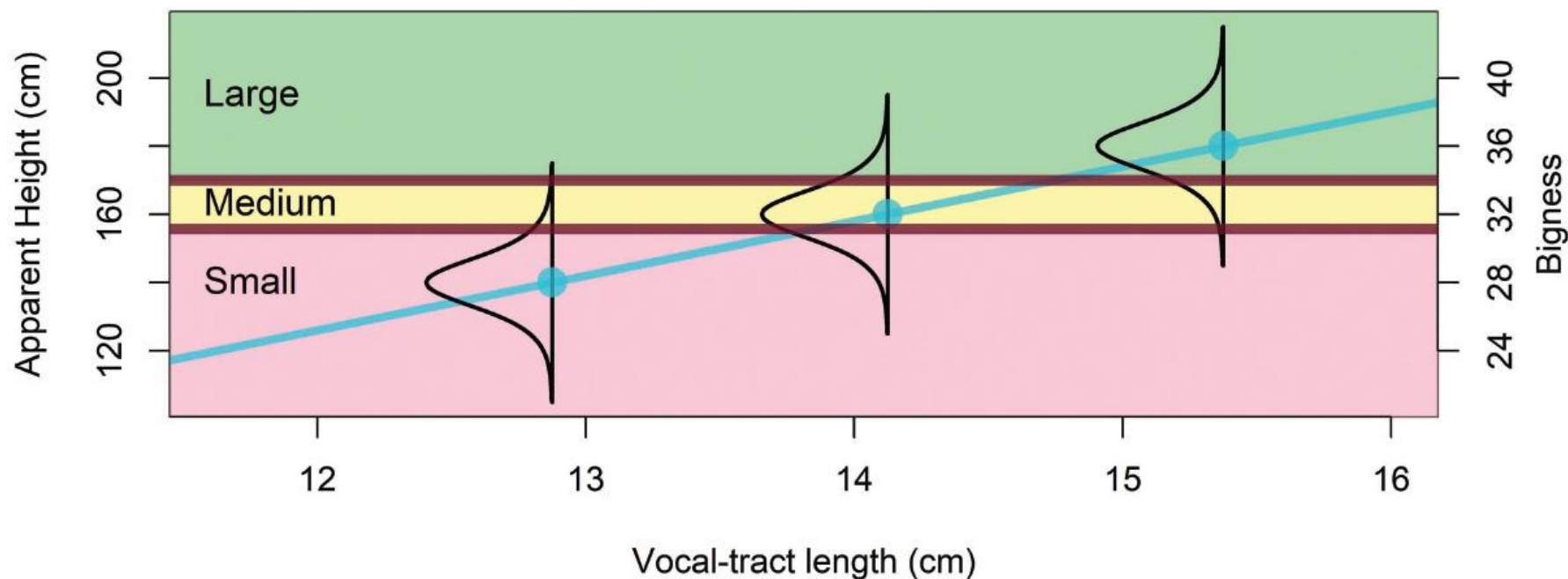
Ordinal Data

- Por razones pedagógicas, vamos a tratar nuestros grupos como categorías ordinales.
- Cambios en VTL cambian la categoría percibida.



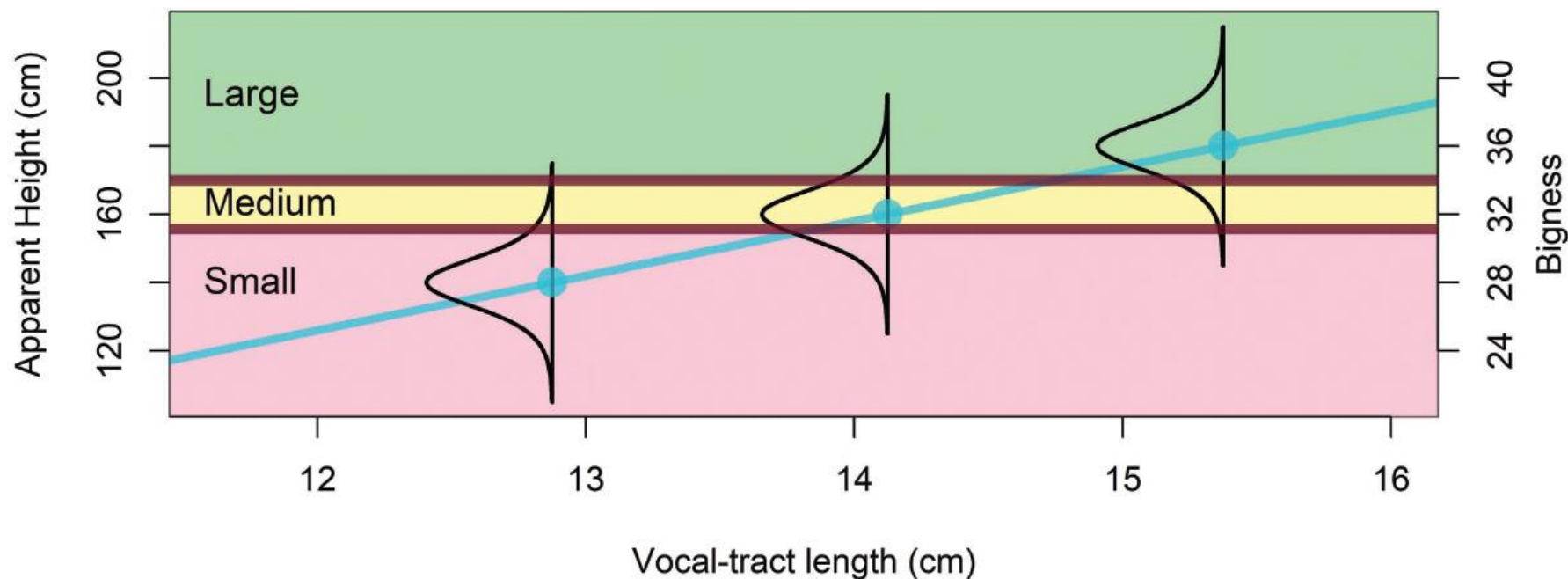
Latent Variable

- Imaginemos en lugar de altura, una variable latente relacionada con la 'grandeza'.
- Dada una VTL, una voz tiene cierta 'grandeza' esperada.
- Pero, también hay variación aleatoria en la percepción.

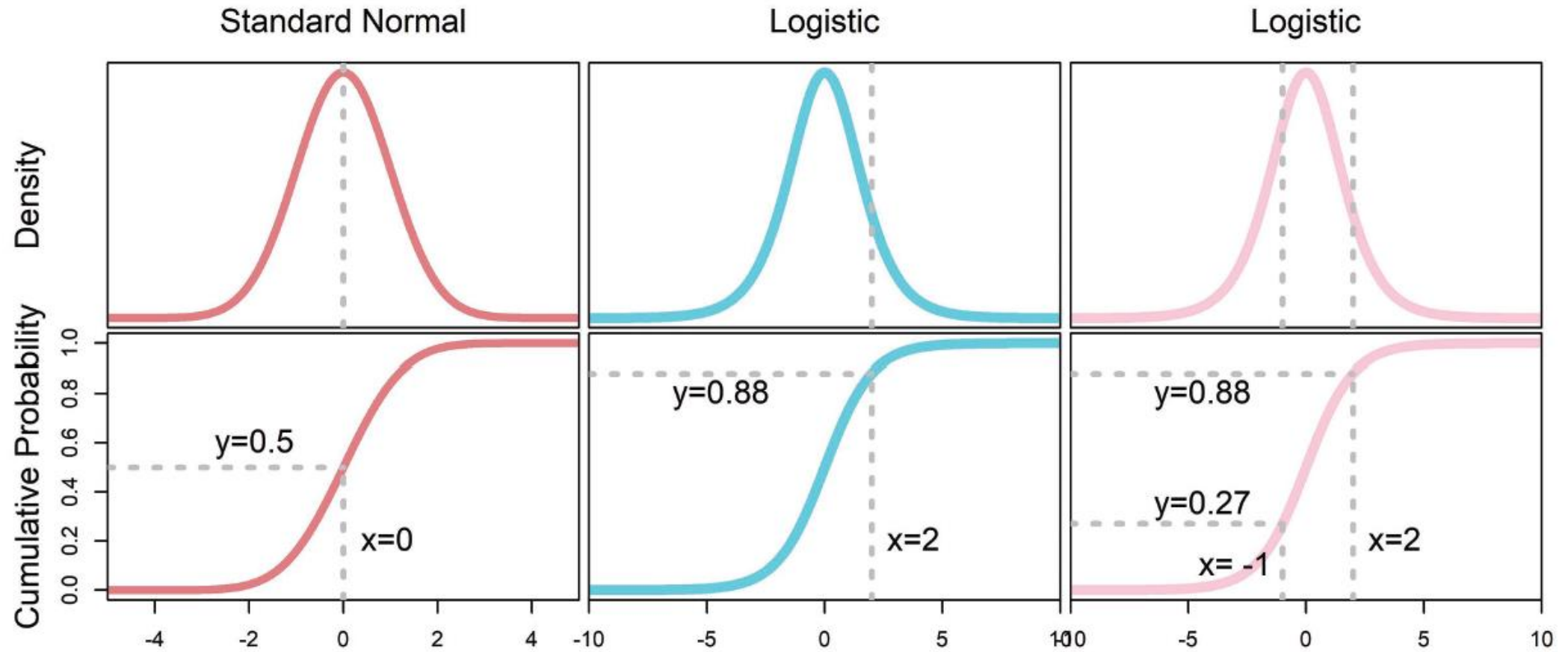


Latent Variable

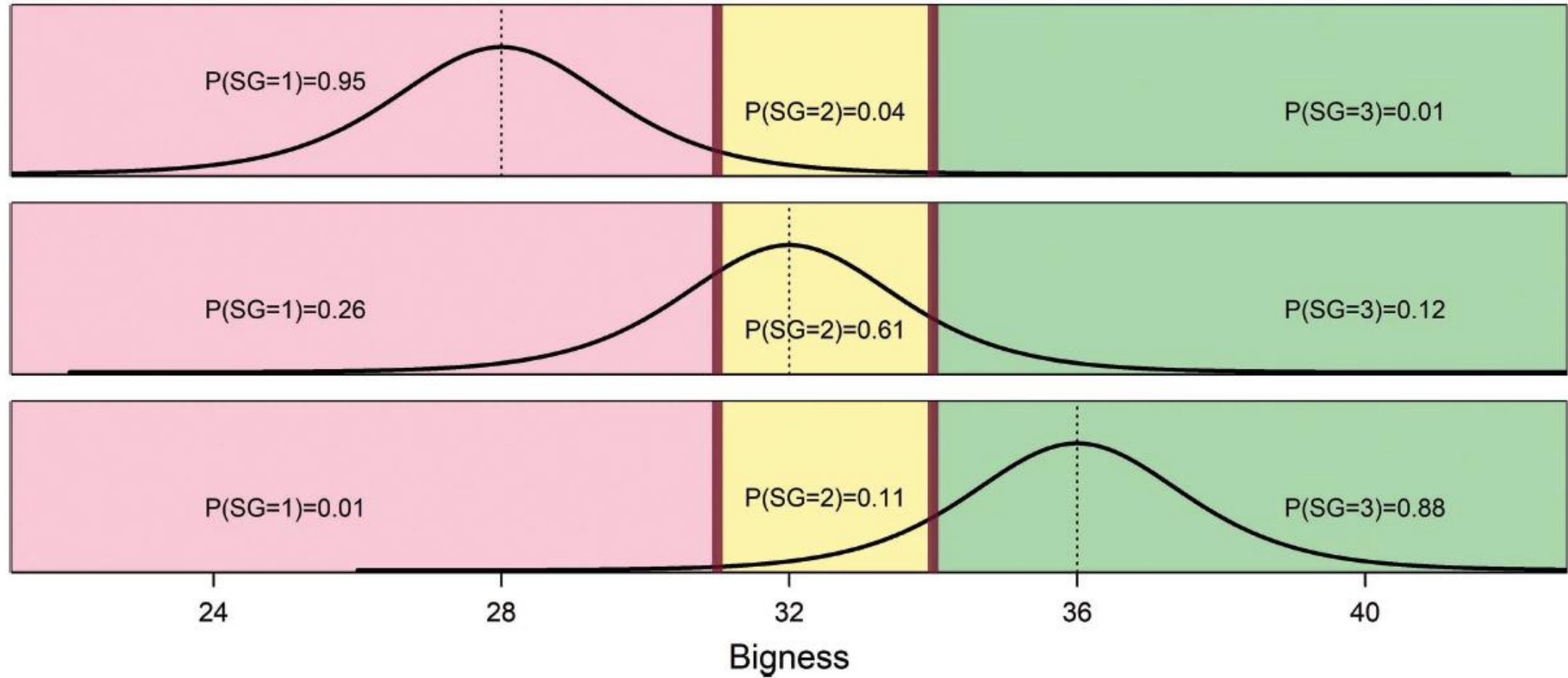
- Asumimos que hay umbrales entre las categorías, y estimamos estos.
- La variable de resultado depende de dónde se encuentre la variable latente en relación con los umbrales.



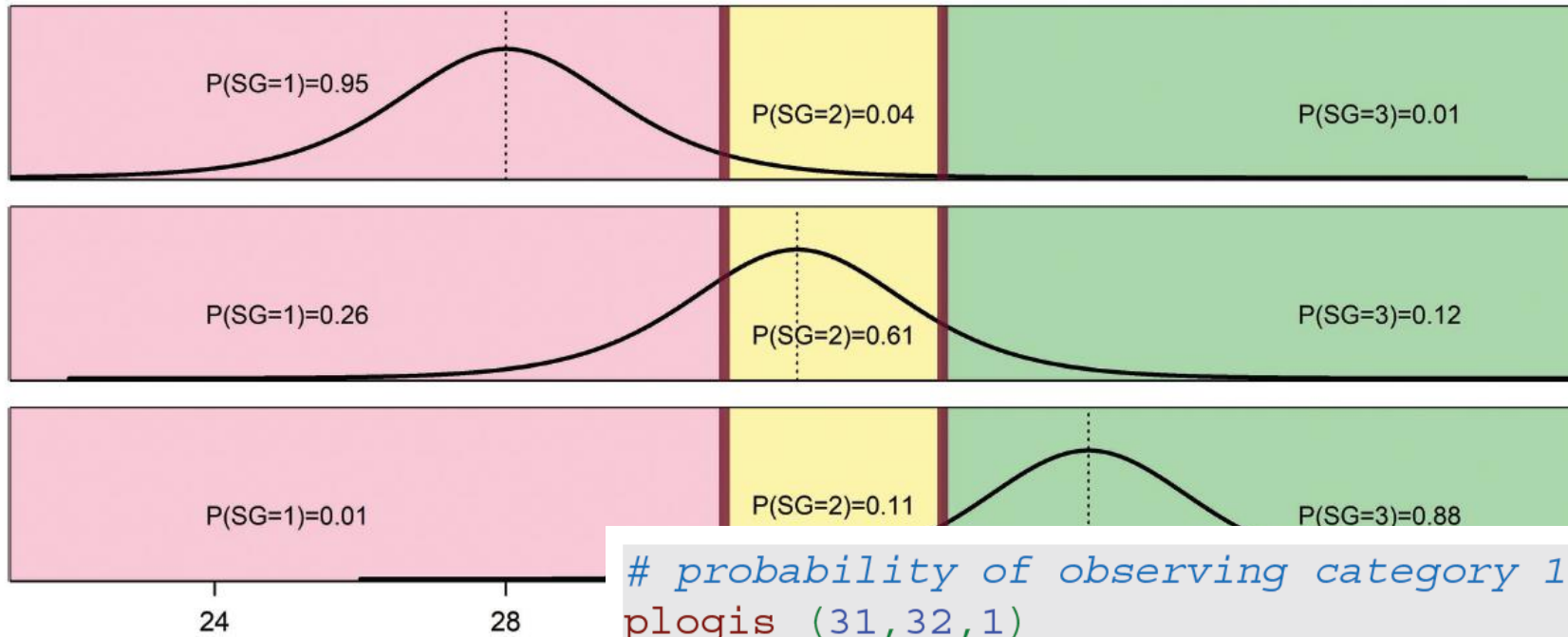
Cumulative Distribution



Probability of Outcomes



Probability of Outcomes



```
# probability of observing category 1
plogis (31,32,1)
## [1] 0.2689

# probability of observing category 2
plogis (34,32,1) - plogis (31,32,1)
## [1] 0.6119

# probability of observing category 3
1 - plogis (34,32,1)
## [1] 0.1192
```

Data and Research Questions

(Q1) Can we predict size group responses for a speaker given their f0 and VTL?

```
library (brms)
library (bmmmb)
data (exp_data)
# new dependent variable: Size Group
SG = 0
SG[exp_data$C=='g'] = 1
SG[exp_data$C=='b'] = 1
SG[exp_data$C=='w'] = 2
SG[exp_data$C=='m'] = 3
exp_data$SG = SG

# preparation of quantitative predictors
# as in previous chapters
exp_data$vtl_original = exp_data$vtl
exp_data$vtl = exp_data$vtl - mean (exp_data$vtl)

exp_data$f0_original = exp_data$f0
exp_data$f0 = exp_data$f0 - mean(exp_data$f0)
exp_data$f0 = exp_data$f0 / 100
```

Description of the Model

$$SG \sim \text{vtl} + f0 + (\text{vtl} + f0 \mid L) + (1 \mid S)$$

$$SG_{[i]} = \begin{cases} 1 & \text{if } z_{[i]} \leq \theta_{[1]} \\ 2 & \text{if } \theta_{[1]} < z_{[i]} \leq \theta_{[2]} \\ 3 & \text{if } \theta_{[2]} \leq z_{[i]} \end{cases}$$

$$z_{[i]} \sim \text{Logistic}(\mu_{[i]}, 1)$$

$$\mu_{[i]} = a_{[i]} + b_{[i]} \cdot \text{vtl}_{[i]} + c_{[i]} \cdot f0_{[i]}$$

$$a_{[i]} = L_{[L_{[i]}]}$$

$$b_{[i]} = VTL + VTL : L_{[L_{[i]}]}$$

$$c_{[i]} = F0 + F0 : L_{[L_{[i]}]}$$

Fitting the Model

```
# Fit the model yourself
```

```
model_ordinal =
```

```
  brms::brm (SG ~ vtl+f0 + (vtl+f0|L) + (1|S), data=exp_data,  
            family="cumulative", chains=4, cores=4, warmup=1000,  
            iter = 5000, thin = 4,  
            prior = c(set_prior("student_t(3, 0, 3)", class = "Intercept"),  
                      set_prior("student_t(3, 0, 3)", class = "b"),  
                      set_prior("student_t(3, 0, 3)", class = "sd"),  
                      set_prior("lkj_corr_cholesky (2)", class = "cor")))
```

```
# Or download it from the GitHub page:
```

```
model_ordinal = bmmmb::get_model ("12_model_ordinal.RDS")
```

Interpreting the Model

```
bmbb::short_summary(model_ordinal)
## Formula:  SG ~ vt1 + f0 + (vt1 + f0 | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      0.54      0.15    0.30    0.89
## sd(vt1)             0.45      0.20    0.07    0.89
## sd(f0)              0.67      0.40    0.04    1.55
## cor(Intercept,vt1)  0.07      0.33   -0.58    0.67
## cor(Intercept,f0)   0.14      0.36   -0.58    0.79
## cor(vt1,f0)         -0.11     0.37   -0.78    0.62
##
## ~S (Number of levels: 139)
##
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      1.27      0.14    1.02    1.57
##
## Population-Level Effects:
##
##           Estimate Est.Error 1-95% CI u-95% CI
## Intercept[1]      -1.86      0.22   -2.32   -1.44
## Intercept[2]       2.58      0.24    2.13    3.07
## vt1                3.16      0.29    2.63    3.76
## f0                 -1.78      0.60   -2.98   -0.58
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI
## disc          1         0         1         1
```

Latent Variable Prediction

```
# make latent variable predictions
predictions_latent = fitted (model_ordinal, scale="linear",
re_formula = NA)
```



```
# inspect first 6 predictions
head (predictions_latent)
```

##		Estimate	Est.Error	Q2.5	Q97.5
##	[1,]	-5.000	0.3558	-5.742	-4.330
##	[2,]	-4.545	0.3958	-5.330	-3.783
##	[3,]	-5.864	0.4669	-6.817	-4.994
##	[4,]	-5.645	0.3486	-6.376	-4.978
##	[5,]	-4.728	0.2945	-5.340	-4.171
##	[6,]	-7.115	0.5044	-8.168	-6.163

Calculating the Outcome Probability

```
# get fixed effects
fixed_effects = brms::fixef(model_ordinal)
```

```
threshold_1 = fixed_effects[1,1]
threshold_2 = fixed_effects[2,1]
vtl_slope = fixed_effects[3,1]
f0_slope = fixed_effects[4,1]
```

```
# get expected values
```

```
mu = exp_data$vtl * vtl_slope + exp_data$f0 * f0_slope
```

```
predictions_manual =
  cbind (round (plogis (threshold_1, mu),5),
        round ((plogis (threshold_2, mu)-plogis
(threshold_1, mu)),5),
        round (1-plogis (threshold_2, mu), 5))
```

```
head (predictions_manual)
```

```
##           [,1]      [,2]      [,3]
## [1,] 0.95834 0.04115 0.00051
## [2,] 0.93592 0.06328 0.00080
## [3,] 0.98201 0.01778 0.00021
## [4,] 0.97772 0.02202 0.00027
## [5,] 0.94604 0.05329 0.00067
## [6,] 0.99478 0.00515 0.00006
```

Outcome Variable Prediction

```
# predict probability of category membership  
predictions_latent = fitted (model_ordinal, re_formula = NA)
```

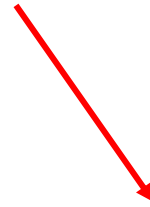
```
# see first six for the first category  
head (predictions_latent[, , 1])  
##           Estimate Est.Error    Q2.5   Q97.5  
## [1,]      0.9557    0.016006  0.9184  0.9803  
## [2,]      0.9314    0.026561  0.8714  0.9714  
## [3,]      0.9799    0.009882  0.9566  0.9933  
## [4,]      0.9763    0.008675  0.9561  0.9896  
## [5,]      0.9435    0.017720  0.9035  0.9723  
## [6,]      0.9941    0.003051  0.9866  0.9981
```

Outcome Variable Prediction

```
# see first six for each category
head (predictions_latent[,1,])
##          P(Y = 1) P(Y = 2)  P(Y = 3)
## [1,]      0.9557 0.043713 5.615e-04
## [2,]      0.9314 0.067657 8.960e-04
## [3,]      0.9799 0.019885 2.480e-04
## [4,]      0.9763 0.023454 2.943e-04
## [5,]      0.9435 0.055775 7.227e-04
## [6,]      0.9941 0.005824 7.232e-05
```

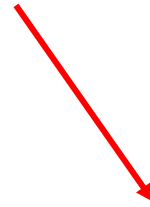

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